THE MICROPROCESSORS & MICROCONTROLLERS

Instructor: MSc. The Tung Than

PRACTICE EXERCISE #3:
USING INTERRUPT

I. Student preparation

- Understanding how to configure and use interrupts.

II. Practice content (5 points)

1. Present and draw a flowchart to handle 2 buttons with the following functions (2 points):
    - Button A: Pause/Resume stopwatch
    - Button B: Reset the stopwatch.
2. Using AT89C51/AT89C52 in combination with 4 7-Segment LED modules and 2 buttons above, design a Sport clock circuit with the ability to count accurately to 1% of seconds, counting range from 00.00 seconds to 99.99 seconds and has 2 buttons to control Pause/Resume and Reset. (3 points)

III. Exercises (5 points)

3. Add 2 buttons to the Sport watch with the following functions:
    a. Button C: Increase the number of seconds counting to 1 second (1 point)
    b. Button D: Decrease the number of seconds to 1 second (1 point)
    c. Button E: Switch the stopwatch speed between normal speed (1×) and double speed (2×). (1 point)
4. When the stopwatch reaches 99.99 seconds, it automatically stops and a LED connected to P1.7 turns on. Pressing the Reset button clears the counter and returns the display to 00.00 seconds. (1 point)
5. Compare the difference between edge interrupt and level interrupt? (1 point)

IV. Report

Compress design files and report files into a file named as follows:

[<LAB...>]-[<Student code>]-Full name

The required report file contains the following contents:

1. Design result (screenshot and pasted in the report).
2. Explain the operating principle of the effects, accompanied by a video (send a Google Drive link) to demonstrate the circuit operation in case the instructor cannot run the design file.
3. Exercise report.